St. Eligius Clinical Interface

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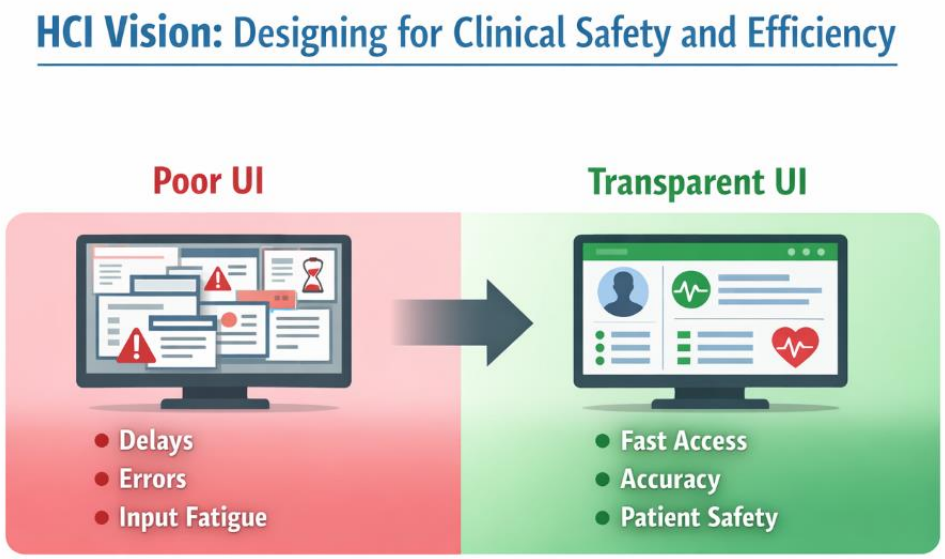
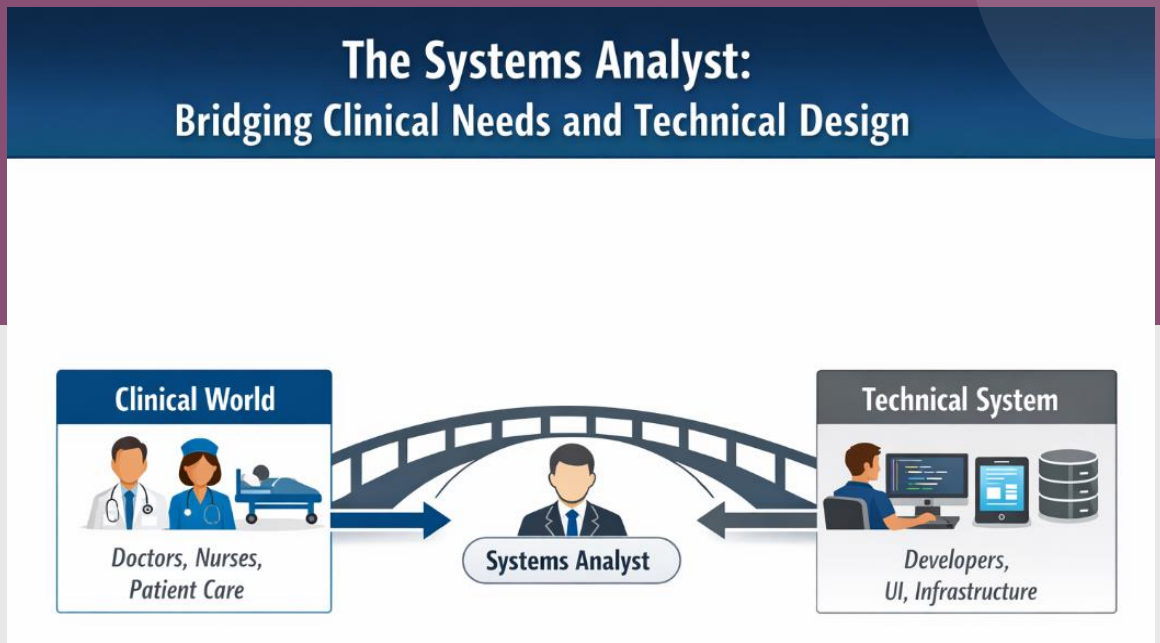
MGMT 480: Systems Analysis & Design

HCI Vision: Designing for Clinical Safety and Efficiency

- Human-Computer Interaction (HCI) = how clinicians interact with the system
- In healthcare, interface design directly impacts patient outcomes
- Poor design → input fatigue, delays, and medical errors
- Goal: transparent interface that disappears into clinical workflow
- Outcome: faster decisions, reduced errors, improved patient safety



The Systems Analyst: Bridging Clinical Needs and Technical Design

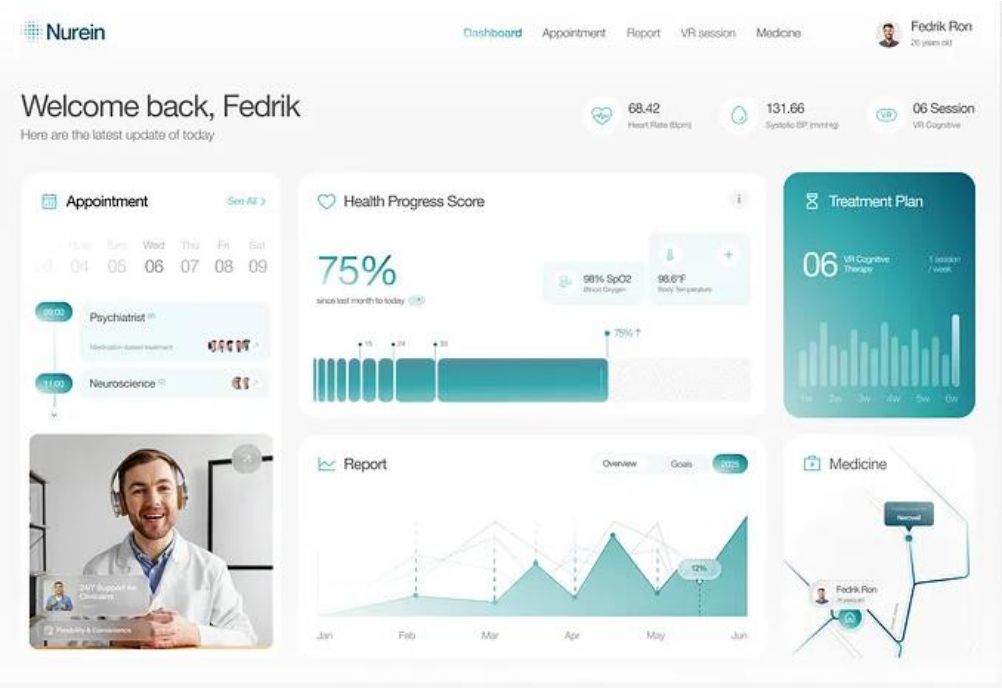


- Systems Analyst = link between clinicians and developers
- Translates clinical workflows → interface requirements
- Ensures system supports real hospital operations
- Aligns usability (HCI) with technical feasibility
- Uses prototyping to identify design risks early
- Goal: safe, efficient, and usable clinical interface

Seven Habits Strategy: Designing for Clinical Use

Understand	Think	Focus on
Understand the Business → high-pressure clinical environment	Think Like a User → minimize clicks, reduce cognitive load	Focus on Usability → fast, intuitive patient data access

Interface Guidelines: Ensuring Usability & Efficiency



Provide Feedback

- System confirms actions (Vitals Updated Successfully)
- Visual indicators (loading spinner, alerts)

Minimize Data Entry

- Auto-fill patient data from records
- Dropdowns and checkboxes replace manual typing

Focus on Task

- Screen displays only critical clinical data
- Eliminates unnecessary fields and distractions

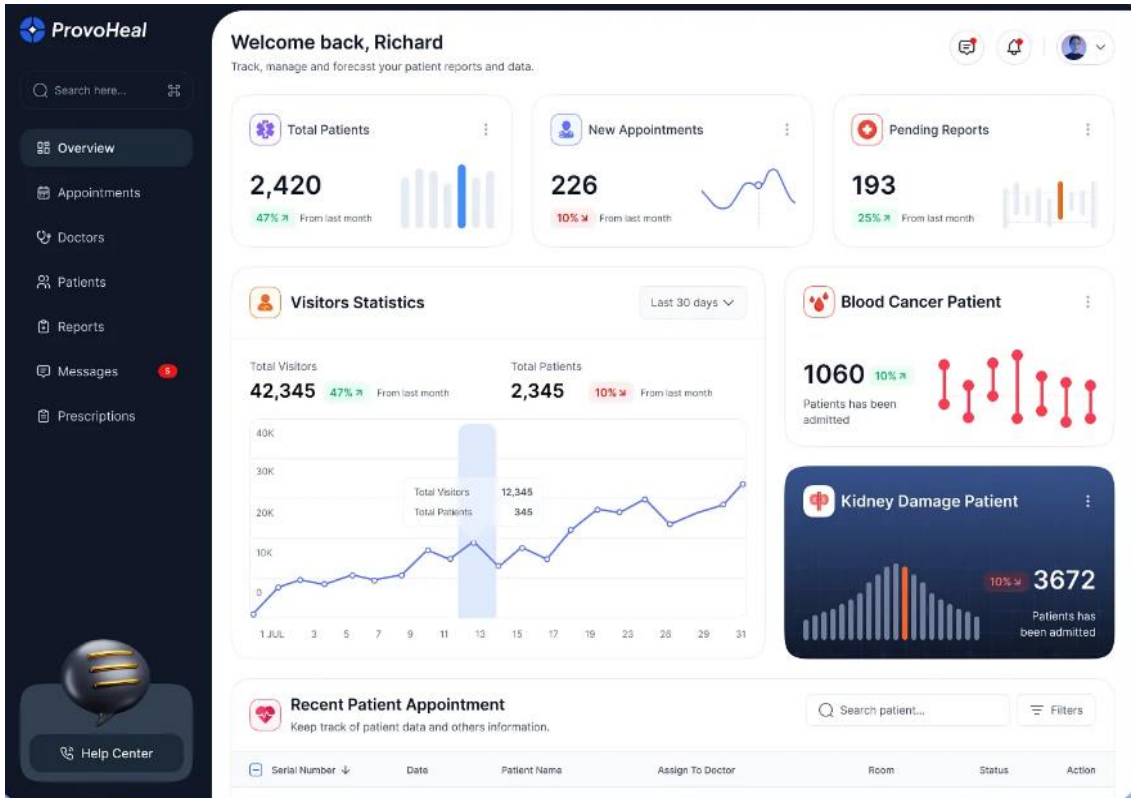
Consistency

- Standardized buttons, labels, and layout across all screens
- Same workflow for all patient interactions

Error Prevention & Handling

- Input validation (e.g., dosage limits)
- Clear error messages with correction guidance

Opening Screen Storyboard: Clinical Dashboard



Patient Search & Selection

Critical Patient Data (Always Visible):

- Blood Type
- Active Allergies
- Current Vital Signs

Quick Actions (Methods):

- update_Vitals()
- calculate_Medication_Dosage()

Real-Time Alerts Panel

3-Second Access to Critical Data

Data Capture & Input: Preventing Clinical Errors

Input Validation Controls

- Enforces safe ranges (dosage limits)

Structured Data Entry

- Dropdowns, checkboxes, predefined fields

Auto-Populated Data

- Pulls existing patient information automatically

Real-Time Error Feedback

- Immediate alerts for incorrect entries

System-Level Safeguards

- Cross-checks allergies before medication entry

Output Strategy: Delivering Actionable Information

Real-Time Clinical Outputs

- Patient summaries and live vitals updates

Exception Reports

- Alerts for abnormal vitals and medication conflicts

Discharge & Summary Reports

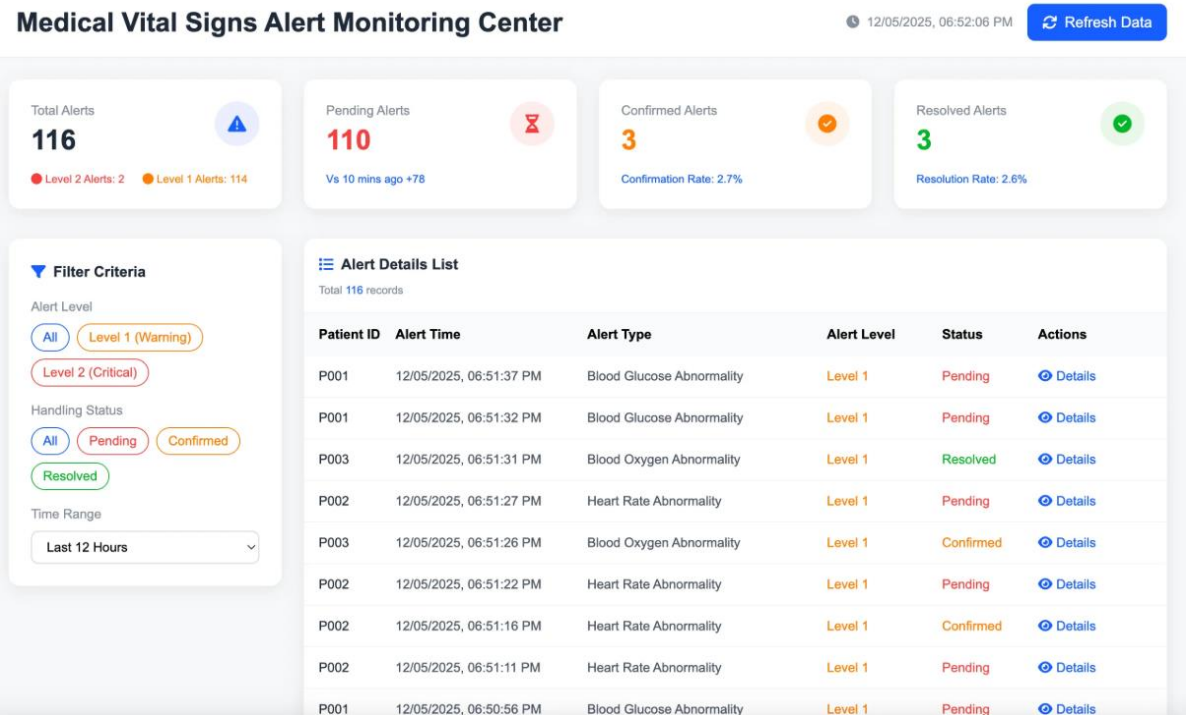
- Auto-generated patient care summaries

Executive-Level Reporting

- Throughput, efficiency, and system performance metrics

Clear & Readable Format

- High-contrast, easy-to-interpret reports



Modular & Responsive Design: One System, Any Device

Responsive interface across devices
- Desktop: full dashboards and data entry
- Tablet: simplified, high-priority patient data
- Mobile: alerts and quick updates

Supports clinical workflow
- Desktop → monitoring and reporting
- Tablet → bedside patient interaction
- Eliminates switching between systems

Technical foundation
- Web-based systems enable multi-device access and scalability
- Modular structure allows easy system expansion

Consistent user experience
- Same system logic across all devices
- Reduces learning curve for staff

Operational efficiency
- Faster access to critical patient data
- Reduces time spent navigating systems
- Improves decision-making speed

Business value
- Increased staff productivity
- Lower training costs
- Scalable for future hospital technologies

Security & Control: Protecting Patient Information

Role-Based Access Control (RBAC)

- Doctors, nurses, admins see only relevant data

Authentication & Authorization

- Secure login + multi-factor authentication

Data Visibility Controls

- Sensitive attributes restricted by user role
- (insurance vs clinical data)

Encapsulation of Patient Data

- Data + logic stored within Patient_Record
- Prevents unauthorized data manipulation

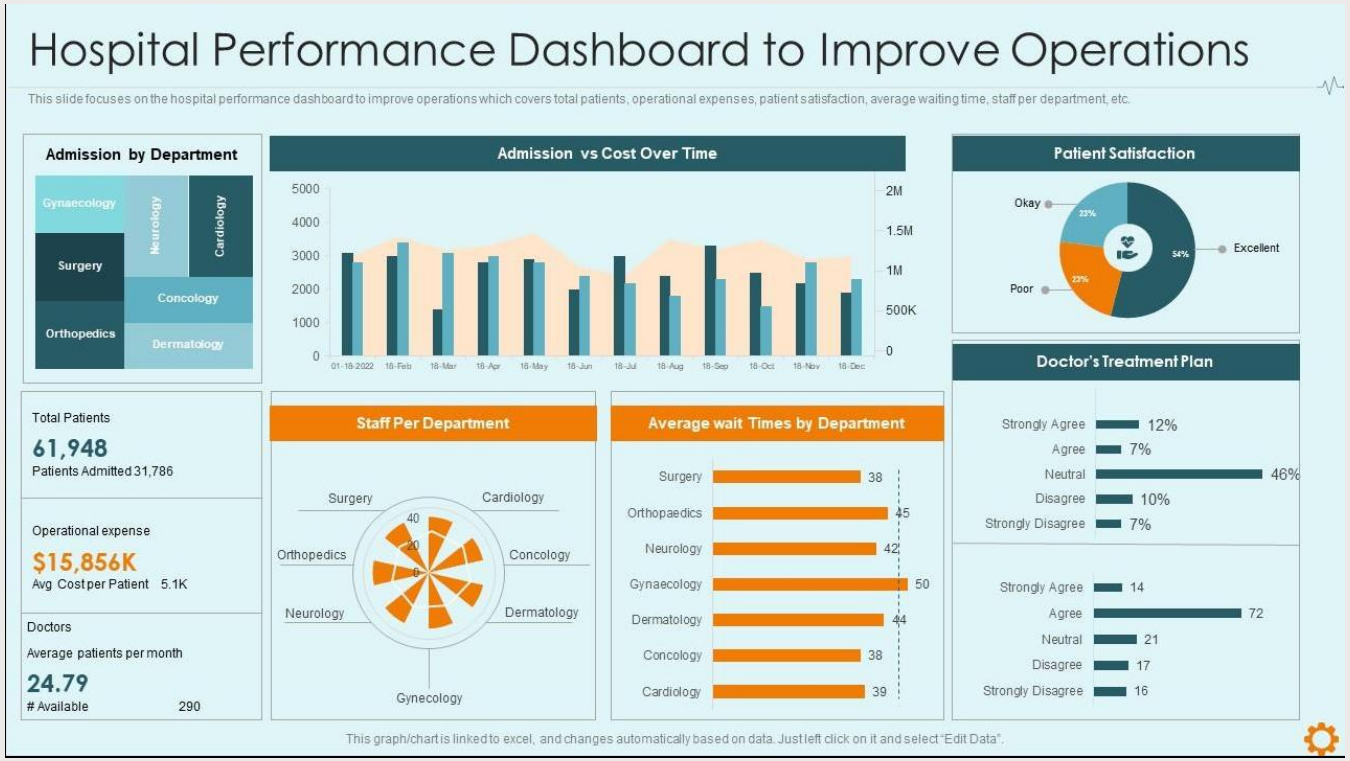
Audit Trails & Monitoring

- Tracks who accessed or modified records

Session Timeouts & Auto-Lock

- Prevents unauthorized access on shared devices

The Bottom Line: Business Impact of Interface Design



Reduced Training Time

- Intuitive design lowers onboarding requirements

Lower Maintenance Costs

- Modular design simplifies updates and fixes

Increased Staff Productivity

- Faster data entry and retrieval

Improved Patient Throughput

- Reduced delays in accessing patient information

Error Reduction = Cost Savings

- Fewer medical errors and liability risks

Scalable System Design

- Supports future expansion without redesign

Questions?

Chief Medical Officer (CMO)

"Dr. Jones, in a trauma situation, which three patient attributes must be visible on the home screen without a single click?"

Head of Nursing

"Does this menu layout match the physical flow of your staff as they move from the nurse's station to the patient bedside?"

Chief Financial Officer (CFO)

"If we reduce data entry time by 15 seconds per patient, what is the projected impact on our daily patient throughput?"

General Staff

"Looking at this storyboard, is there any terminology here that feels like 'computer talk' rather than 'hospital talk?"

Answers!

Chief Medical Officer (CMO)

Three attributes must be visible without interacting with the patient: their active allergies, vital signs, and blood type. Allergies ensure the patient does not receive potentially lethal medications. Vitals provide an instant view of the patient's condition. Knowing the blood type is essential for transfusions in emergency situations.

Head of Nursing

The menu layout allows nurses to navigate from the nurse's station to the patient's bedside. At the nurse's station, they can access the patient's records. At the patient's bedside, they can access the patient's vital signs, view their allergies, and update their records.

Chief Financial Officer (CFO)

Reducing the amount of time it takes to enter data by 15 seconds per patient will save the hospital time. With 100 patients per day, the hospital will save 25 minutes of data entry per day. With 300 patients per day, the hospital will save 75 minutes of data entry per day. This will increase the patients that can be treated and the efficiency of the nurses and the operational capacity of the hospital.

General Staff

The system avoids technical terms and utilizes language commonly used in the hospital environment. Patient_Record is displayed as Patient Chart and update_Vitals() is displayed as Update Vitals. This reduces confusion and reduces the amount of time to train the staff on the functions of the interface.

References/Sources

ChatGPT (Used to create the visual images)

St. Eligius Systems Modernization Project Brief. MGMT 480 – Systems Analysis & Design, course handout.

Designing St. Eligius Clinical Interface Assignment. *MGMT 480 – Systems Analysis & Design*.

Systems Analyst UI Design Approach. *MGMT 480 Course Material*.

Tilley, S. (2020). *Systems Analysis and Design* (12th ed.). Cengage Learning.